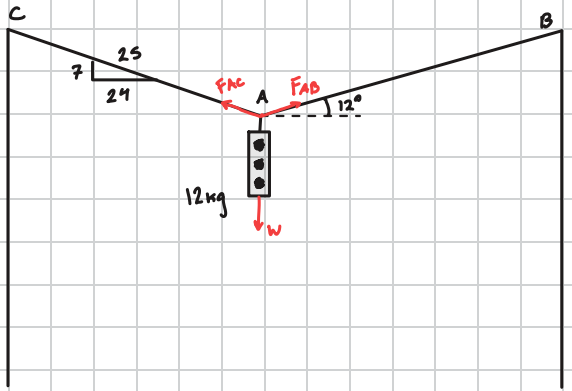
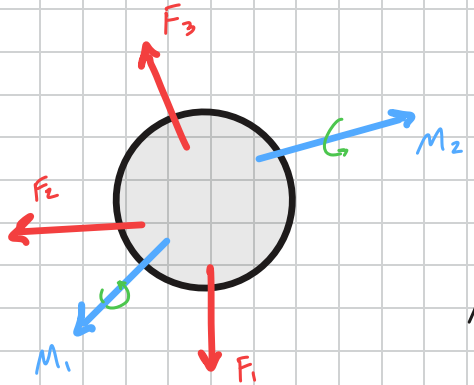
Week 3

*Don't worry about equilibrium

Equilibrium



Rigid body equilibrium

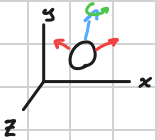


No more 3D cuz it's out of our scope

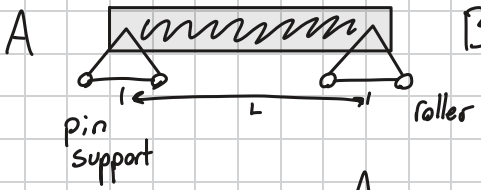
rigid body is solved in 2D

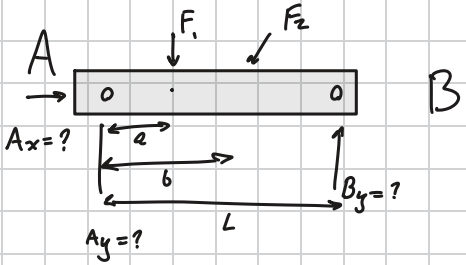
$\Sigma F_x \quad \Sigma F_y$

ΣM_z



Most cases start with moments then move to forces





$\Sigma F_x = 0$

$\Sigma F_y = 0$

$\Sigma M_z = 0$

$\Sigma M_{oz} = 0$ $\Sigma M_{pz} = 0$

one of the force equations

$\Sigma F_x = 0$ or $\Sigma F_y = 0$

$\Sigma M_{OA} = 0$

$A_x(0) + A_y(0) - F_1(a) - F_2(b) + B_y(L) = 0$

$B_y = ?$

if the unknown are more than the system :3

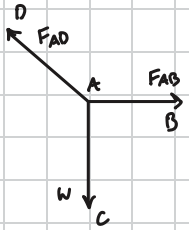
basicly more forces than needed like support cables on a bridge

Statically indeterminate

FBD

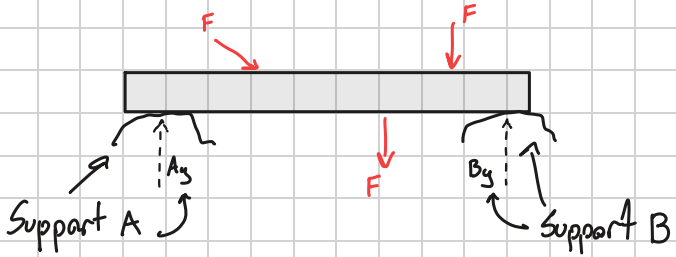
Huge part of statics

2D Free body diagrams



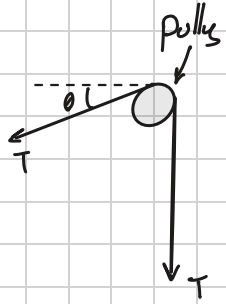
$\Sigma F_x = 0$

$\Sigma F_y = 0$



if support stops a translation or rotation it has to be stated properly

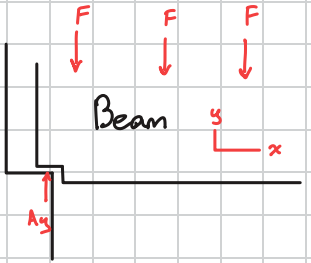
Cables, wires, pulleys

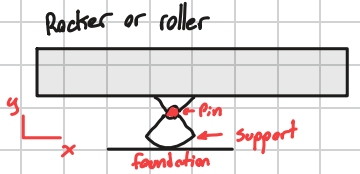


reglect weight of string/pulley

spring can be under tension

$F = kx$ or $F = kS$





Free movment

FBD



force isn't always ∥ to y